SUPERR SELECTOR

SEASON 2 | ROUND 1

Most Reliable Batters and Bowlers Under Pressure

Forensic data audit • Transparent scoring • Rajasthan Royals 2027 strategy

Prepared 13 November 2026

Executive summary

This report presents two independently calculated ranking populations. The official Round 1 submission tables use the IPL (2022–2026) and T20I population specified in the brief. A second expanded research analysis adds the supplied Emerging Teams and Rising Stars T20 competitions. Pressure is operationalised through early-innings collapse and the death phase. Players are compared with context-matched peers and with their own ordinary baseline; small samples are shrunk toward peer rates and match-to-match consistency is included.

Decision item	Primary result
Eligible player pool	96 batters; 156 bowlers
Primary data scope	IPL 2022–2026 + T20 Internationals
Delivery-level sample	194,014 records across 837 matches
Pressure opportunities	50,874 unique delivery records
Primary score	0–100 relative composite; not a probability
Expanded strategy	Retrospective RR 2026 squad audit + conditional IPL 2027 auction plan

Dual-scope decision: the IPL + T20I tables remain the official Round 1 submission rankings because they follow the competition population specified in the brief. The expanded IPL + T20I + Emerging Teams + Rising Stars analysis is presented as a major, separately rerun research result. It identifies the effect of broader competition coverage without silently changing the official submission population.

Interpret the order as the output of one explicit model, not a claim that a player is universally the best under every pressure situation. The sensitivity section shows how rankings move when reasonable modelling choices change.

1. Dataset scope and coverage

The supplied source pack contains six CSV files. The primary selection rule retains competitions whose format is T20I or whose competition name exactly matches IPL 20XX. The filter is applied from competition metadata and match IDs before any context, pressure, baseline or ranking calculation.

Coverage item	Observed value
All source match rows	883
All source ball-by-ball rows	200,999
Primary-scope match rows	853
Primary-scope matches with ball data	837
Primary-scope matches without ball data	16
Primary-scope deliveries	194,014
Primary-scope competition IDs	136 (5 IPL; 131 T20I with matches)
IPL matches, 2022–2026	371 total; 367 with ball-by-ball
T20I matches	482 total; 470 with ball-by-ball
Additional expanded-scope T20 competitions	2 competitions; 30 matches
Date range in primary matches	22 Jan 2022 – 1 Nov 2026

Sixteen primary-scope matches have no corresponding delivery rows (4 IPL and 12 T20I). They remain counted in match coverage but cannot contribute ball-by-ball metrics. The 141-row competition metadata file includes 134 T20I competition entries; 131 T20I competition IDs are represented in the selected matches.

The two additional T20 competitions are Mens T20 Emerging Teams Asia Cup 2024 (15 matches) and Asia Cup Rising Stars 2025 (15 matches). All 30 matches have ball-by-ball records, adding 6,985 delivery rows to the expanded run. They are valid supplied T20 data; their separate treatment is a scope decision, not a data-quality rejection.

Expanded-scope sensitivity: effect on the Top 10

Measure	Official: IPL + T20I	Expanded research: all supplied T20 + T20I
Matches / matches with ball data	853 / 837	883 / 867
Delivery records	194,014	200,999
Batter Top 10 overlap	Primary reference	9/10
Bowler Top 10 overlap	Primary reference	8/10

Batter changes: Liam Livingstone enters; Ramandeep Singh leaves. Bowler changes: Ehsan Khan enters; Rasikh Salam enters; Ben Dwarshuis leaves; Rahul Chahar leaves. The shared players can also move in rank and score; the expanded run recomputes the peer-relative model and eligible candidate pool over the larger population. The expanded lists are a separately calculated research result; the official IPL + T20I ranking CSVs remain unchanged for the Unstop submission.

Interpretation: this comparison demonstrates that competition selection changes the candidate pool, peer population and final ranks. It does not establish that the expanded ranking is inherently more accurate. The official Unstop tables remain IPL + T20I; the expanded tables are a substantial additional research result and are shown separately on the next page.

2. Expanded research ranking: Emerging Teams + Rising Stars

This is the separately rerun expanded population: IPL 2022–2026, T20 Internationals, Men's T20 Emerging Teams Asia Cup 2024 and Asia Cup Rising Stars 2025. The two additional competitions contribute 30 matches and 6,985 deliveries. The expanded peer baselines, eligible candidate pool, performance components and consistency components are recalculated; these are not primary-scope scores copied into a larger table.

Expanded Top 10 batters

Rank	Player	Expanded index
1	Najibullah Zadran	87.73
2	Sikandar Raza	85.13
3	Rinku Singh	81.77
4	Tristan Stubbs	81.12
5	Naman Dhir	80.51
6	Nurul Hasan	78.82
7	Rishabh Pant	76.96
8	Glenn Phillips	74.93
9	Babar Azam	74.85
10	Liam Livingstone	73.42

Expanded Top 10 bowlers

Rank	Player	Expanded index
1	Sunil Narine	85.88
2	Ehsan Khan	85.43
3	Hasan Mahmud	83.28
4	Rashid Khan	82.41
5	Dasun Shanaka	80.51
6	Akeal Hosein	80.45
7	Rasikh Salam	79.76
8	Riley Meredith	79.48
9	Adam Zampa	78.90
10	Nandre Burger	78.42

Interpretation and use: these expanded rankings are the primary result for the project's broader research question, while the IPL + T20I rankings remain the official Round 1 submission result. Do not mix scores across scopes: each score is relative to its own recalculated candidate and peer population. A rank or score change after expansion reflects a changed reference population, not necessarily a change in the player's underlying ability.

2. Data integrity and field-level decisions

Check	Observed result	Treatment
Delivery ordering	194,014 rows; 0 duplicate delivery numbers; 0 innings with gaps	Pass: use innings_ball_num for ordering
Innings sequence starts	0 groups not starting at ball 1	Pass
Wicket state	Monotonic; max 9 wickets before a delivery	Count wickets before the current ball
Regular / additional innings	193,988 regular; 26 additional	Innings 3+ retained but not pressure-rated
Batting balls faced	186,540	Exclude wides; include Overstep; retain 1 Penalty row
Wide / Overstep records	7,474 / 565	Wide excluded; Overstep counted as faced / excluded from bowler candidate legal balls
Boundary flag anomalies	20 four-run rows and 3 six-run rows have no matching boundary flag	Use supplied flags; do not infer missing flags
Dismissal attribution	10,486 wicket rows; 1 missing dismissed_player_id	Attribute by dismissed_player_id; do not guess missing ID

Why the run fields are labelled conservatively

The source `runs` column sums to 261,119 in the primary dataset. However, the extras fields do not provide complete run amounts: for example, 505 byes records and 2,661 leg-byes records have zero in the source runs field. Consequently, batting rates use the label recorded_runs and bowling economy / true dot-ball rates are not produced.

Of 10,486 wicket records, 9,905 are treated as bowler-creditable (Bowled, Caught, LBW, Stump, Hit Wicket); 581 are not. Run-outs are not credited to bowlers. The 25 bowler-creditable stumpings on wides remain credited, as dismissal credit is separate from ball legality.

3. Pressure definition

3.1 Early-collapse pressure

An innings enters early-collapse pressure if at least five wickets have fallen by the end of the first 50% of its allocated length. Pressure begins on the delivery after the fifth wicket falls. The fifth-wicket delivery itself is not counted as post-collapse pressure. In a standard 20-over innings, the cutoff is the end of over 10.

3.2 Death-phase pressure

The final 25% of allocated innings length is treated as the death phase. In a standard 20-over innings this is overs 16–20. The rule applies regardless of match closeness or wickets in hand. Shortened-innings boundaries scale with the allocated innings length, rounded to full overs.

3.3 Role-specific pressure

Batters are ranked in the de-duplicated union: early-collapse OR death phase. Bowlers are ranked on death-over context only. A batting-side collapse does not automatically mean the bowling side is under pressure; early-collapse bowling is retained as supporting analysis, not used as the bowler's primary pressure category.

Pressure event	Delivery records
Early-collapse	11,096
Death phase	43,600
Overlap	3,822
Unique combined pressure	50,874
Early-collapse innings	175 across 173 matches

The rule is intentionally observable and reproducible. It does not claim to capture every pressure state. The source fields cannot reliably reconstruct required run rate, target, full team totals or match closeness; those unobservable states are not fabricated or included.

4. Metrics and peer baselines

4.1 Batting metrics

Metric	Calculation	Direction / weight
Recorded runs rate	recorded_runs / balls_faced × 100	Higher is better · 50%
Dismissal rate	dismissals attributed by dismissed_player_id / balls_faced × 100	Lower is better · 30%
Boundary rate	(is_four + is_six flags) / balls_faced × 100	Higher is better · 20%

4.2 Bowling metrics

Metric	Calculation	Direction / weight
Bowler wicket rate	Bowler-creditable wickets / candidate legal balls × 100	Higher is better · 60%
Flagged boundary rate conceded	(four + six flags) conceded / candidate legal balls × 100	Lower is better · 40%

Candidate legal bowling balls exclude `wide` and `Overstep`; the single Penalty-coded record is retained in the primary denominator because its event semantics are not documented. A penalty-excluded variant differs by one ball. Economy rate and true dot-ball rate are deliberately omitted because the extras fields do not support them.

4.3 Peer comparison

- Each player is compared with a leave-one-player-out, opportunity-weighted peer rate; the subject player is excluded from the comparator.
- Peer-context hierarchy: competition + innings + phase; then format + season + innings + phase; then format + innings + phase.
- Competition-level peer support requires at least 300 other-player opportunities and 8 other players; broader fallback requires at least 50 opportunities and 3 players.
- Phase boundaries use the same proportional allocation rule: powerplay = first 30%, death = final 25%, middle = intervening overs. This prevents comparing raw powerplay and death rates as though they were the same context.

Baseline coverage is 99.76% for batter context rows and 99.75% for bowler context rows. Unmatched rows are not assigned invented peer rates.

5. Reliability adjustment and scoring

5.1 Shrinkage

For a player with observed numerator x, opportunity count n and a matched peer rate r (per 100 opportunities), the sample-adjusted rate is ((x + k × r / 100) / (n + k)) × 100. The primary player-level prior k is 60 opportunities; the sensitivity grid tests 30 and 100. Each player-match rate is similarly shrunk toward its matched peer using 12 prior opportunities.

This is a transparent fixed-prior shrinkage method, not a fitted empirical-Bayes model. Its purpose is to reduce extreme rates from small samples while retaining the observed signal.

5.2 Composite score

Component	Construction
Performance (70%)	For each metric: 50% percentile of sample-adjusted pressure level vs peer + 50% percentile of pressure uplift vs the player's ordinary peer-relative standing; then apply metric weights.
Consistency (30%)	For each metric: 50% positive adjusted-match share percentile + 30% median adjusted match-performance percentile + 20% inverse dispersion percentile; then apply metric weights.
Final index	0.70 × performance component + 0.30 × consistency component; relative percentile composite on a 0–100 scale.

5.3 Eligibility before ranking

Role	Pressure sample	Ordinary comparison	Peer coverage
Batter	≥100 combined-pressure balls and ≥10 pressure matches	≥100 ordinary balls and ≥10 ordinary matches	≥80% for all required metrics
Bowler	≥60 death-over candidate legal balls and ≥10 death matches	≥60 ordinary candidate legal balls and ≥10 ordinary matches	≥80% for all required metrics

These rules leave 96 eligible batters and 156 eligible bowlers. All 20 selected players have 100% baseline coverage in the required metrics. The score is an ordering aid; 87.62 does not mean an 87.62% probability of succeeding under pressure.

6. Primary top 10 — batters under pressure

#	Player	Score	Pressure balls	Matches	Positive matches	Mean uplift
1	Najibullah Zadran	87.62	127	12	66.7%	+37.4 pp
2	Sikandar Raza	85.09	231	19	56.1%	+32.4 pp
3	Naman Dhir	82.69	211	24	62.5%	+31.9 pp
4	Rinku Singh	81.71	605	62	59.7%	+28.2 pp
5	Tristan Stubbs	80.79	463	45	57.8%	+29.6 pp
6	Nurul Hasan	78.35	124	15	62.2%	+41.0 pp
7	Rishabh Pant	76.31	249	27	58.0%	+21.3 pp
8	Ramandeep Singh	75.19	106	18	61.1%	+43.1 pp
9	Babar Azam	74.93	111	12	61.1%	+15.6 pp
10	Glenn Phillips	74.67	251	26	55.1%	+16.0 pp

How to read the list

Najibullah Zadran leads with 127 combined-pressure balls across 12 matches. Sikandar Raza has a larger sample (231 balls, 19 matches) and a slightly higher performance component, but a lower consistency component; the final score is a 70/30 blend. Naman Dhir is third with 211 balls across 24 matches. The rank is driven by the declared composite, not by reputation or a manual override.

The closeness of scores matters: Rinku Singh (81.71) and Tristan Stubbs (80.79) are separated by less than one point. The model does not justify treating that gap as a decisive cricketing difference. Use sample volume and role fit alongside the index.

Candidate evidence

The table shows opportunity volume, distinct pressure matches, the share of baseline-supported matches with positive peer-adjusted results, and mean pressure-versus-ordinary uplift. A positive share is not the same as a win rate; it is a share of match-level metric observations with a positive adjusted peer-relative result.

7. Primary top 10 — bowlers under pressure

#	Player	Score	Death balls	Matches	Positive matches	Mean uplift
1	Sunil Narine	85.97	216	35	68.6%	+24.5 pp
2	Hasan Mahmud	83.82	108	12	62.5%	+25.3 pp
3	Rashid Khan	82.54	540	80	63.8%	+22.1 pp
4	Akeal Hosein	80.98	108	18	69.4%	+18.5 pp
5	Dasun Shanaka	80.29	110	18	58.3%	+34.9 pp
6	Riley Meredith	79.43	118	13	61.5%	+44.9 pp
7	Adam Zampa	79.16	240	38	64.5%	+18.9 pp
8	Nandre Burger	78.76	87	14	57.1%	+18.5 pp
9	Ben Dwarshuis	77.60	91	13	61.5%	+27.6 pp
10	Rahul Chahar	76.67	122	18	52.8%	+25.5 pp

How to read the list

Sunil Narine leads with 216 death-over candidate legal balls across 35 matches. Rashid Khan has the largest sample among the top three (540 balls across 80 matches), while Hasan Mahmud ranks second with 108 balls across 12 matches. The difference is based on peer-adjusted wicket and boundary-flag rates plus match consistency.

The bowling list is more sensitive to metric choice than the batting list. Under equal metric weights, only 7 of the primary 10 bowlers remain. This is a material limitation and should be disclosed in any judge presentation, not hidden behind the main score.

Boundary flags are used as supplied. The model cannot fully reconstruct bowler-charged runs, so the bowling score is not an economy-rate ranking.

8. Robustness and sensitivity

8.1 Pressure threshold

Collapse cutoff	Eligible batters	Eligible bowlers	Batter overlap	Bowler overlap
By over 8	95	156	9/10	10/10
By over 12	94	156	7/10	9/10

At an 8-over collapse cutoff, 9/10 batter and 10/10 bowler places overlap the primary list. At 12 overs, overlap falls to 7/10 and 9/10. The batter ranking is therefore more dependent on the early-collapse boundary than the bowler ranking.

8.2 Prior and sample-threshold sensitivity

Scenario	Role	Eligible candidates	Top-10 overlap
prior30_bat100_bowl60_matches10	batters	96	10/10
prior30_bat100_bowl60_matches10	bowlers	156	9/10
prior100_bat100_bowl60_matches10	batters	96	9/10
prior100_bat100_bowl60_matches10	bowlers	156	9/10
prior60_bat80_bowl40_matches10	batters	110	10/10
prior60_bat80_bowl40_matches10	bowlers	156	10/10
prior60_bat120_bowl80_matches10	batters	81	7/10
prior60_bat120_bowl80_matches10	bowlers	141	10/10
prior60_bat100_bowl60_matches8	batters	97	10/10
prior60_bat100_bowl60_matches8	bowlers	173	7/10

8.3 Metric-weight sensitivity

Weight scenario	Batter overlap	Bowler overlap
primary	10/10	10/10
equal_metrics	10/10	7/10
runs_and_wickets_emphasis	9/10	9/10
survival_and_boundary_control	9/10	7/10
boundary_emphasis	10/10	9/10
consistency_heavier	10/10	9/10
performance_heavier	10/10	9/10

The selected weights are reasoned modelling choices, not objectively optimal constants. If a judge prefers different weights, the alternative lists in `metric_weight_sensitivity.json` show the exact player-level results. Under a 120/80 opportunity eligibility rule, the batter top-10 overlap falls to 7/10; with an 8-match rule, the bowler overlap falls to 7/10.

Shortened-innings audit: 134 second-innings delivery rows across two matches inherit the first-innings revised allocation when the second-innings field is blank and the observed second innings reaches the same over cap. Disabling this inheritance changes the unique pressure count from 50,874 to 50,880 but leaves all 20 top-10 names unchanged.

9. Rajasthan Royals: 2026 squad and role audit

This section applies the Round 1 pressure-performance findings to Rajasthan Royals' IPL 2027 squad requirements, connecting the statistical rankings with role suitability and conditional recruitment decisions. It uses the supplied IPL 2026 squad mapping and player appearances, then checks the role structure against Rajasthan Royals' official 2026 squad page. The analysis is retrospective: it does not assume the 2026 squad will remain intact for IPL 2027.

2026 RR batting output	Matches	Recorded runs	Balls faced	Source cross-check
Vaibhav Sooryavanshi	16	776	327	Official RR season page: 776
Dhruv Jurel	16	515	333	Official RR season page: 515
Yashasvi Jaiswal	16	427	280	Official RR season page: 427
Donovan Ferreira	15	317	177	Official RR squad page: 317
Riyan Parag	14	309	196	Official RR squad page: 309
Ravindra Jadeja	14	266	197	Official RR squad page: 266

The supplied ball-by-ball `runs` totals for these six 2026 RR batters match the published figures on the official RR squad page. This is a targeted cross-check, not proof that every player and every extras record in the full dataset is error-free.

Observed bowling deployment in 2026

Bowler	Matches	PP legal balls	Middle legal balls	Death legal balls	Credited wkts
Jofra Archer	16	204	48	108	25
Brijesh Sharma	13	73	108	96	14
Nandre Burger	13	156	54	51	14
Ravi Bishnoi	9	12	120	24	11
Ravindra Jadeja	14	0	198	0	10
Yash Punja	8	6	156	18	9
Sandeep Sharma	6	48	18	52	5
Tushar Deshpande	7	48	42	48	4

The deployment data shows multiple death-over options (Archer, Brijesh, Nandre, Sandeep and Tushar) and substantial middle-over spin from Jadeja, Bishnoi and Yash Punja. Therefore, a blanket claim that RR lacks death bowling or spin is not supported by its 2026 usage. The more defensible question is which of these players remain retained and available, and whether the role mix is resilient to absence/injury.

10. RR fit review — ranked batters

The statistical list and recruitment shortlist are different things. The table below reviews each batter on role fit against the 2026 RR structure. Franchise labels refer to published 2026 squad information, updated where a 2027 trade was officially confirmed by 13 November 2026. They do not establish retention, release, registration or availability for the auction.

#	Player	Profile	2026 squad / trade status	RR view
1	Najibullah Zadran	Overseas · Middle-order batter / finisher	Not listed in reviewed 2026 IPL squad list	Watchlist only; consider if RR lose a finisher.
2	Sikandar Raza	Overseas · Batting all-rounder	Not listed in reviewed 2026 IPL squad list	Flexible overseas all-rounder; overlaps current RR options.
3	Naman Dhir	Indian · Middle-order batter / utility option	Mumbai Indians (2026)	Useful Indian middle-order depth if released by MI.
4	Rinku Singh	Indian · Finisher	Kolkata Knight Riders (2026)	Strong Indian finisher fit; KKR/availability first.
5	Tristan Stubbs	Overseas · Middle-order batter / wicketkeeper	Delhi Capitals (2026)	Quality, but overlaps overseas middle-order/keeper cover.
6	Nurul Hasan	Overseas · Wicketkeeper-batter	Not listed in reviewed 2026 IPL squad list	Low priority: RR already has wicketkeeping cover.
7	Rishabh Pant	Indian · Wicketkeeper-batter / top-middle order	Delhi Capitals (2027 trade confirmed)	Not practical: DC trade confirmed; Jurel is established.
8	Ramandeep Singh	Indian · Batting all-rounder / lower-order option	Kolkata Knight Riders (2026)	Useful Indian lower-order utility if released.
9	Babar Azam	Overseas · Top-order batter	Not listed in reviewed 2026 IPL squad list	Do not shortlist; IPL registration/eligibility unverified.
10	Glenn Phillips	Overseas · Batting all-rounder / wicketkeeper option	Gujarat Titans (2026)	Versatile, but overseas middle-order role is covered.

Interpretation: the most attractive role profiles are Indian middle-order/finishing options (Naman Dhir, Rinku Singh and Ramandeep Singh) because they can address depth without consuming an overseas slot. That is a conditional role-fit assessment only: none is an available target until the official 2027 list confirms release and registration. The top two statistical ranks are not automatically the best recruitment fits.

11. RR fit review — ranked bowlers

The bowler list is likewise screened for role complementarity, not merely score. RR's 2026 deployment already included pace across the powerplay and death, plus middle-over spin. A top-10 bowler should be considered only if the retained squad leaves a specific phase or skill gap.

#	Player	Profile	2026 squad / trade status	RR view
1	Sunil Narine	Overseas · Spin all-rounder	Kolkata Knight Riders (2026)	Elite, but spin/all-rounder overlap; KKR status.
2	Hasan Mahmud	Overseas · Fast bowler	Not listed in reviewed 2026 IPL squad list	Conditional overseas pace option; verify registration/fitness.
3	Rashid Khan	Overseas · Leg-spin all-rounder	Gujarat Titans (2026)	High quality, but spin overlap and GT status/price.
4	Akeal Hosein	Overseas · Left-arm orthodox spinner	Chennai Super Kings (2026)	Left-arm spin overlap with current RR middle-over options.
5	Dasun Shanaka	Overseas · Batting all-rounder / seam option	Rajasthan Royals (2026)	RR 2026 member; assess retention and role-use.
6	Riley Meredith	Overseas · Fast bowler	Not listed in reviewed 2026 IPL squad list	Pace depth only after fitness and release checks.
7	Adam Zampa	Overseas · Leg-spinner	Not listed in reviewed 2026 IPL squad list	Spin contingency only if current depth changes.
8	Nandre Burger	Overseas · Left-arm fast bowler	Rajasthan Royals (2026)	RR 2026 member; assess retention and fitness.
9	Ben Dwarshuis	Overseas · Left-arm fast bowler / lower-order batting	Punjab Kings (2026)	Left-arm seam cover if released; overseas slot needed.
10	Rahul Chahar	Indian · Leg-spinner	Chennai Super Kings (2026)	Indian spin backup only if attrition creates a gap.

For RR, retaining a functional phase-balanced core may be more valuable than buying another highly ranked overseas name. Dasun Shanaka and Nandre Burger are already in the 2026 RR group; they are retention/role decisions, not external targets. Hasan Mahmud, Riley Meredith and other players absent from the reviewed 2026 squad list are not assumed to be registered or available.

12. IPL 2027 auction intelligence and conditional RR plan

As at 12 November 2026, the retention deadline is reported as 17 November 2026. Cricbuzz reports a mini-auction scheduled for 16 December 2026 in Goa, registration closing 23 November, and the final auction register expected on 3 December. Treat the auction date and venue as reported schedule information pending formal BCCI confirmation; player availability is not yet final.

Planning checkpoint	Reported date / status	Practical implication
RR retention/release submission	17 Nov 2026	Do not finalise team gap or purse before this list
Player registration window closes	23 Nov 2026	Registration still does not prove a player is retained/released by a franchise
Final auction register	3 Dec 2026 (reported)	Use this as the definitive purchasable-player universe
Mini-auction	16 Dec 2026, Goa (reported)	Prepare role tiers and price ceilings, subject to confirmed register
2027 total salary cap	₹157 crore	This is not RR's remaining auction purse

Conditional recruitment priorities

- Priority 1 — protect a reliable death-overs pace core. Retain/replace based on actual availability among the 2026 death-usage group; only seek an additional specialist if retentions, injury/fitness or role duplication create a genuine gap.
- Priority 2 — preserve middle-over wicket-taking options. Jadeja, Bishnoi and Yash Punja carried meaningful 2026 middle-over workloads. A new spinner is a succession/contingency decision, not an automatic gap.
- Priority 3 — assess finishing depth after retentions. Ferreira, Hetmyer and Shubham Dubey all recorded 2026 middle/death usage; review role clarity and Indian/overseas balance before bidding for another finisher.
- Priority 4 — maintain an Indian pace succession option if the 2027 retained group leaves limited domestic fast-bowling cover. Set reserve prices only after the released-player list and purse are confirmed.

The 2027 salary-cap rule states ₹157 crore for the total salary cap. It must not be presented as the franchise's auction purse: the remaining purse depends on salaries retained, releases, trades and applicable rules. The final team-wise auction balance should be recalculated from the confirmed retention list and contracts.

No named player is labelled a confirmed RR target or an available auction player in this report. The statistical top-10 lists are performance evidence, not availability evidence. Existing franchise contract, registration, overseas-player rules, final auction register, fitness, price and squad-slot constraints must all be verified before naming a target.

13. Technical validation and reproducibility

The project is implemented as separate Python modules, with a reproducible command-line pipeline and unit tests. The final recorded automated test run passed 57 tests. The scope-comparison tests check stable-ID matching, overlap and entry/exit identification, duplicate detection, ten-player requirements and excluded-competition ball-by-ball coverage.

Regression checks covered

- The delivery taking the fifth wicket is not treated as the first post-collapse pressure delivery; the next delivery is.
- Death phase uses allocated innings length; a standard 20-over innings starts death pressure in over 16.
- Overlapping early-collapse/death records are counted once in the combined batting pressure set.
- Run-outs and other non-bowler dismissals do not count as bowler-creditable wickets.
- Wides and Overstep are excluded from candidate legal bowling balls; sample eligibility is enforced.
- Competition scope includes only IPL 2022+ and T20I by default; expanded T20 scope is explicit.
- Sensitivity options recalculate rankings; the pressure-ranking stability flag no longer silently does nothing.
- Primary and expanded competition-scope runs are independently executed; expanded outputs are stored separately and cannot overwrite the primary rankings.

Reproduce from the six supplied CSVs

Place the six source CSVs in the project's `data/` folder, install the version-constrained requirements, then run:

```
python -m pip install -r requirements.txt
python -m pytest -q
python main.py --audit-context --audit-pressure --audit-batting --audit-bowling --audit-baselines --audit-reliability
--audit-rankings --audit-pressure-ranking-stability --audit-rr-squad
python build_supplemental_outputs.py
python run_scope_comparison.py
python generate_report.py
```

Judge-ready explanations

- Why not required run rate? The source pack does not contain complete extras/total fields; required runs and full team totals cannot be safely reconstructed.
- Why 60-opportunity prior? It is a declared shrinkage choice; 30 and 100 are tested, and the result changes modestly but not identically.
- Why death only for bowlers? Early batting collapse is pressure on the batter; it does not automatically make the bowler's task high-pressure.
- Why are weights not optimal? They encode the project's chosen balance among run output, survival, boundary scoring, wicket taking and boundary suppression; sensitivity tables disclose the consequences of alternatives.

14. Limitations, interpretation and final recommendation

- The pressure model is an observable proxy, not a complete model of psychological or match-state pressure. It does not use required run rate, wickets in hand, target, or match closeness.
- The revised-overs field has no data dictionary. Values are interpreted as ball allocations because supplied values are multiples of six; the two narrow second-innings inheritance cases are separately audited.
- The source runs/extras fields are incomplete for full score reconciliation. Batting metrics are named recorded_runs; bowling economy and true dot-ball rates are excluded.
- Twenty-three deliveries with 4 or 6 recorded runs lack the matching boundary flag. The analysis uses flags as supplied rather than inferring missing events.
- Fixed priors, score weights, cutoffs and eligibility thresholds are modelling decisions. Sensitivity shows material ranking changes, particularly for bowlers under equal metric weights and batters under a 12-over cutoff.
- The 2027 squad, final player register and RR auction purse were not known at the 12 November 2026 cut-off. Recruitment advice is conditional and role-based.
- The 0–100 score is a relative composite percentile, not probability, confidence or a guarantee of future performance.

Final recommendation

Use the strict IPL+T20I top-10 lists as the primary Round 1 submission, retain the model audit files with the report, and defend the pressure rule as a clear operational definition rather than as the only possible form of cricket pressure. Disclose the sensitivity results prominently. For RR 2027, wait for confirmed retentions and the final auction register before naming purchasable targets or calculating the available purse.

External references for the RR and 2027 planning appendix

- Rajasthan Royals, IPL 2026 squad and player roles/stats: https://www.rajasthanroyals.com/rr-players (accessed 13 November 2026).
- Rajasthan Royals, official IPL 2026 season statistics: https://www.rajasthanroyals.com/stats (accessed 13 November 2026).
- BBC Sport, IPL 2026 squads (published 16 December 2025; updated 27 March 2026): https://www.bbc.co.uk/sport/cricket/articles/cp891x46p65o (accessed 13 November 2026).
- IPL/BCCI, Rishabh Pant and Kuldeep Yadav complete DC–LSG trade (23 June 2026): https://www.iplt20.com/news/article/rishabh-pant-and-kuldeep-yadav-complete-high-profile-trade-between-dc-and-lsg (accessed 13 November 2026).
- IPL/BCCI, CSK–SRH four-player trade (5 November 2026): https://www.iplt20.com/news/article/csk-and-srh-agree-to-four-player-trade (accessed 13 November 2026).
- IPL/BCCI, TATA IPL 2026 player retentions announced: https://www.iplt20.com/news/article/tata-ipl-2026-player-retentions-announced (accessed 13 November 2026).
- IPL/BCCI, TATA IPL Player Regulations 2025–27: https://www.iplt20.com/news/article/ipl-governing-council-announces-tata-ipl-player-regulations-2025-27 (accessed 13 November 2026).
- Cricbuzz, BCCI extends IPL retention deadline to November 17; mini-auction schedule reported for 16 December in Goa: https://www.cricbuzz.com/cricket-news/140762/bcci-extends-ipl-retention-deadline-to-november-17-cricbuzzcom (published 2 November 2026; accessed 12 November 2026).

Data provenance: supplied competition and match CSVs; derived tables and audit JSONs are generated by this project's code. External sources above support only the RR squad and 2027 rules/calendar context.

Appendix A. Worked scoring examples

The calculations below use the actual primary-run candidate-score rows. Rates are shown per 100 opportunities. The peer rate is leave-one-player-out and opportunity-weighted within the selected context cohort. The peer baseline estimates what comparable players produced in the same competition/innings/phase context; leaving the subject player out prevents their own performance from inflating their comparator. Shrinkage then pulls a player's observed rate toward that peer rate by the declared 60-opportunity prior, reducing extreme small-sample estimates without giving extra score points merely for having more balls. For dismissal and boundary-conceded rates, lower is better; relative advantage is direction-adjusted.

A1. Batter example — Najibullah Zadran

Input	Observed value
Recorded runs / pressure balls	247 / 127
Raw recorded-runs rate	247 ÷ 127 × 100 = 194.49
Matched peer rate	151.41 recorded runs per 100 balls
Prior	60 opportunities at the peer rate
Shrunk rate	(247 + 60 × 1.5141) ÷ (127 + 60) × 100 = 180.67
Adjusted relative advantage	(180.67 ÷ 151.41 − 1) × 100 = +19.32%

For the runs metric, Najibullah's pressure-level percentile is 92.71 and his pressure-uplift percentile is 100.00; their 50/50 average is 96.35. Applying the 50% runs weight contributes 48.18 points to the performance component. The same operation is performed for dismissals (30%) and boundary flags (20%). The resulting performance component is 91.04; consistency is 79.65; final index = 0.70 × 91.04 + 0.30 × 79.65 = 87.62.

A2. Bowler example — Hasan Mahmud

Input	Observed value
Bowler-creditable wickets / death balls	12 / 108
Raw wicket rate	12 ÷ 108 × 100 = 11.11
Matched peer wicket rate	8.39 per 100 candidate legal balls
Shrunk wicket rate	(12 + 60 × 0.083904) ÷ (108 + 60) × 100 = 10.14
Adjusted relative advantage	(10.14 ÷ 8.39 − 1) × 100 = +20.85%
Boundary control	18 flags / 108 balls; peer 20.16; shrunk rate 17.91 = 11.14% lower than peer

Hasan's wicket and boundary performance percentiles are 88.46 and 82.37. Applying the 60/40 weights gives a performance component of 86.03. His consistency component is 78.67; final index = 0.70 × 86.03 + 0.30 × 78.67 = 83.82.

A3. Why the two close comparisons resolve this way

Comparison	Player A	Player B	Model explanation
Batter	Najibullah: 87.62; P 91.04; C 79.65; 127 balls / 12 matches	Rinku: 81.71; P 86.09; C 71.47; 605 balls / 62 matches	Rinku has much more evidence volume and a slightly higher mean peer-relative pressure result, but Najibullah scores higher on the model's combined pressure-level/uplift percentiles and match-consistency component. Sample size is an eligibility/evidence safeguard, not a bonus that automatically raises rank.
Bowler	Hasan: 83.82; P 86.03; C 78.67; 108 balls / 12 matches	Rashid: 82.54; P 84.29; C 78.44; 540 balls / 80 matches	Rashid has a stronger wicket-rate peer percentile, but his boundary metric has negative pressure uplift (−2.48 pp). Hasan's more balanced wicket/boundary pressure-plus-uplift composite produces the higher performance component. Their consistency components are nearly equal.

These explanations describe this model's ranking logic, not a claim that the lower-ranked player is categorically inferior. The differences are conditional on the declared pressure definition, peer cohorts, weights and eligibility rules.

Appendix B. Player-by-player evidence — batters

Each selected batter's row shows the evidence behind inclusion: pressure opportunity volume, match count, share of positive peer-adjusted match observations, average pressure standing versus peers, uplift from the player's ordinary peer-relative standing, and the two score components. Evidence-volume bands are descriptive only: higher volume = at least 250 pressure balls and 20 matches; moderate volume = at least 150 balls and 15 matches; otherwise qualified smaller sample. They are not confidence levels or probabilities.

#	Batter	Pressure balls / matches	Positive match share	Pressure vs peers	Uplift vs ordinary	P / C	Index	Evidence volume
1	Najibullah Zadran	127 / 12	66.7%	+23.1%	+37.4 pp	91.0 / 79.7	87.62	Qualified; smaller sample
2	Sikandar Raza	231 / 19	56.1%	+27.0%	+32.4 pp	92.2 / 68.5	85.09	Moderate volume
3	Naman Dhir	211 / 24	62.5%	+23.5%	+31.9 pp	86.8 / 73.0	82.69	Moderate volume
4	Rinku Singh	605 / 62	59.7%	+27.5%	+28.2 pp	86.1 / 71.5	81.71	Higher volume
5	Tristan Stubbs	463 / 45	57.8%	+22.5%	+29.6 pp	84.5 / 72.0	80.79	Higher volume
6	Nurul Hasan	124 / 15	62.2%	+16.1%	+41.0 pp	75.1 / 86.0	78.35	Qualified; smaller sample
7	Rishabh Pant	249 / 27	58.0%	+19.9%	+21.3 pp	77.2 / 74.1	76.31	Moderate volume
8	Ramandeep Singh	106 / 18	61.1%	+7.8%	+43.1 pp	73.7 / 78.7	75.19	Qualified; smaller sample
9	Babar Azam	111 / 12	61.1%	+22.7%	+15.6 pp	72.7 / 80.1	74.93	Qualified; smaller sample
10	Glenn Phillips	251 / 26	55.1%	+23.1%	+16.0 pp	79.6 / 63.1	74.67	Higher volume

How to interpret the individual profiles

- Najibullah Zadran (#1): 127 pressure balls in 12 matches; 66.7% positive match share; mean pressure-relative result +23.1%; mean uplift +37.4 pp. Performance 91.04, consistency 79.65, final index 87.62. Evidence volume: qualified; smaller sample.
- Sikandar Raza (#2): 231 pressure balls in 19 matches; 56.1% positive match share; mean pressure-relative result +27.0%; mean uplift +32.4 pp. Performance 92.19, consistency 68.53, final index 85.09. Evidence volume: moderate volume.
- Naman Dhir (#3): 211 pressure balls in 24 matches; 62.5% positive match share; mean pressure-relative result +23.5%; mean uplift +31.9 pp. Performance 86.82, consistency 73.04, final index 82.69. Evidence volume: moderate volume.
- Rinku Singh (#4): 605 pressure balls in 62 matches; 59.7% positive match share; mean pressure-relative result +27.5%; mean uplift +28.2 pp. Performance 86.09, consistency 71.47, final index 81.71. Evidence volume: higher volume.
- Tristan Stubbs (#5): 463 pressure balls in 45 matches; 57.8% positive match share; mean pressure-relative result +22.5%; mean uplift +29.6 pp. Performance 84.53, consistency 72.05, final index 80.79. Evidence volume: higher volume.
- Nurul Hasan (#6): 124 pressure balls in 15 matches; 62.2% positive match share; mean pressure-relative result +16.1%; mean uplift +41.0 pp. Performance 75.05, consistency 86.04, final index 78.35. Evidence volume: qualified; smaller sample.
- Rishabh Pant (#7): 249 pressure balls in 27 matches; 58.0% positive match share; mean pressure-relative result +19.9%; mean uplift +21.3 pp. Performance 77.24, consistency 74.14, final index 76.31. Evidence volume: moderate volume.
- Ramandeep Singh (#8): 106 pressure balls in 18 matches; 61.1% positive match share; mean pressure-relative result +7.8%; mean uplift +43.1 pp. Performance 73.70, consistency 78.66, final index 75.19. Evidence volume: qualified; smaller sample.
- Babar Azam (#9): 111 pressure balls in 12 matches; 61.1% positive match share; mean pressure-relative result +22.7%; mean uplift +15.6 pp. Performance 72.71, consistency 80.10, final index 74.93. Evidence volume: qualified; smaller sample.
- Glenn Phillips (#10): 251 pressure balls in 26 matches; 55.1% positive match share; mean pressure-relative result +23.1%; mean uplift +16.0 pp. Performance 79.64, consistency 63.09, final index 74.67. Evidence volume: higher volume.

Appendix C. Player-by-player evidence — bowlers

Bowler opportunities are death-phase candidate legal balls. Wickets are restricted to the declared bowler-creditable dismissal types; boundary figures use supplied flags. A high pressure-versus-peer result and a high pressure-uplift result are separate signals. Bowler evidence-volume bands are descriptive only: higher volume = at least 200 death balls and 20 matches; moderate volume = at least 100 balls and 12 matches; otherwise qualified smaller sample. They are not confidence levels or probabilities.

#	Bowler	Death balls / matches	Positive match share	Pressure vs peers	Uplift vs ordinary	P / C	Index	Evidence volume
1	Sunil Narine	216 / 35	68.6%	+32.2%	+24.5 pp	86.4 / 84.9	85.97	Higher volume
2	Hasan Mahmud	108 / 12	62.5%	+16.0%	+25.3 pp	86.0 / 78.7	83.82	Moderate volume
3	Rashid Khan	540 / 80	63.8%	+32.1%	+22.1 pp	84.3 / 78.4	82.54	Higher volume
4	Akeal Hosein	108 / 18	69.4%	+17.7%	+18.5 pp	81.0 / 80.9	80.98	Moderate volume
5	Dasun Shanaka	110 / 18	58.3%	+19.0%	+34.9 pp	86.8 / 65.1	80.29	Moderate volume
6	Riley Meredith	118 / 13	61.5%	+12.6%	+44.9 pp	83.6 / 69.7	79.43	Moderate volume
7	Adam Zampa	240 / 38	64.5%	+20.5%	+18.9 pp	80.4 / 76.2	79.16	Higher volume
8	Nandre Burger	87 / 14	57.1%	+11.6%	+18.5 pp	77.5 / 81.7	78.76	Qualified; smaller sample
9	Ben Dwarshuis	91 / 13	61.5%	+18.7%	+27.6 pp	81.7 / 68.0	77.60	Qualified; smaller sample
10	Rahul Chahar	122 / 18	52.8%	+16.0%	+25.5 pp	84.0 / 59.5	76.67	Moderate volume

- Sunil Narine (#1): 216 death balls in 35 matches; 68.6% positive match share; mean pressure-relative result +32.2%; mean uplift +24.5 pp. Performance 86.41, consistency 84.94, final index 85.97. Evidence volume: higher volume.
- Hasan Mahmud (#2): 108 death balls in 12 matches; 62.5% positive match share; mean pressure-relative result +16.0%; mean uplift +25.3 pp. Performance 86.03, consistency 78.67, final index 83.82. Evidence volume: moderate volume.
- Rashid Khan (#3): 540 death balls in 80 matches; 63.8% positive match share; mean pressure-relative result +32.1%; mean uplift +22.1 pp. Performance 84.29, consistency 78.44, final index 82.54. Evidence volume: higher volume.
- Akeal Hosein (#4): 108 death balls in 18 matches; 69.4% positive match share; mean pressure-relative result +17.7%; mean uplift +18.5 pp. Performance 81.03, consistency 80.86, final index 80.98. Evidence volume: moderate volume.
- Dasun Shanaka (#5): 110 death balls in 18 matches; 58.3% positive match share; mean pressure-relative result +19.0%; mean uplift +34.9 pp. Performance 86.79, consistency 65.10, final index 80.29. Evidence volume: moderate volume.
- Riley Meredith (#6): 118 death balls in 13 matches; 61.5% positive match share; mean pressure-relative result +12.6%; mean uplift +44.9 pp. Performance 83.59, consistency 69.74, final index 79.43. Evidence volume: moderate volume.
- Adam Zampa (#7): 240 death balls in 38 matches; 64.5% positive match share; mean pressure-relative result +20.5%; mean uplift +18.9 pp. Performance 80.45, consistency 76.17, final index 79.16. Evidence volume: higher volume.
- Nandre Burger (#8): 87 death balls in 14 matches; 57.1% positive match share; mean pressure-relative result +11.6%; mean uplift +18.5 pp. Performance 77.50, consistency 81.71, final index 78.76. Evidence volume: qualified; smaller sample.
- Ben Dwarshuis (#9): 91 death balls in 13 matches; 61.5% positive match share; mean pressure-relative result +18.7%; mean uplift +27.6 pp. Performance 81.73, consistency 67.96, final index 77.60. Evidence volume: qualified; smaller sample.
- Rahul Chahar (#10): 122 death balls in 18 matches; 52.8% positive match share; mean pressure-relative result +16.0%; mean uplift +25.5 pp. Performance 84.04, consistency 59.47, final index 76.67. Evidence volume: moderate volume.

Appendix D. Match-level performance spotlights

These examples show what a selected player's pressure-labelled segment looked like in a specific match. They are illustrative evidence, not cherry-picked inputs to the ranking: each player's final score uses all eligible pressure and ordinary observations across the dataset. Runs are labelled recorded_runs because the source extras fields do not support full scorecard reconciliation.

D1. Batting pressure segments

Player	Date	Competition	Opponent	Recorded runs / pressure balls	Boundary flags
Najibullah Zadran	2022-08-30	Mens Twenty20 Asia Cup 2022	Bangladesh	42 off 13	1×4, 6×6
Naman Dhir	2024-05-17	IPL 2024	Lucknow Super Giants	57 off 24	3×4, 5×6
Rinku Singh	2026-04-26	IPL 2026	Lucknow Super Giants	56 off 24	4×4, 5×6
Ramandeep Singh	2026-04-14	IPL 2026	Chennai Super Kings	30 off 16	4×4, 1×6

Najibullah's 42 recorded runs from 13 pressure-labelled balls against Bangladesh and Rinku's 56 from 24 against Lucknow are concrete examples of high-output segments. Naman Dhir's 57 from 24 against Lucknow and Ramandeep Singh's 30 from 16 against Chennai show why their middle/lower-order role profiles are relevant to the RR appendix. These isolated innings do not replace the full-sample model.

D2. Bowling death-over segments

Player	Date	Competition	Opponent	Credited wickets / death balls	Boundary flags conceded
Hasan Mahmud	2023-03-09	England in Bangladesh T20I Series 2023	England	2 wickets / 12 legal balls	0 fours, 0 sixes
Rashid Khan	2025-10-31	Afghanistan in Zimbabwe T20I Series 2025	Zimbabwe	3 wickets / 12 legal balls	0 fours, 0 sixes

Hasan Mahmud's two wickets from 12 death-over candidate legal balls against England (9 March 2023) and Rashid Khan's three from 12 against Zimbabwe (31 October 2025) illustrate wicket-taking under the model's death-phase definition. These match examples are consistent with the broader sample evidence but are not the reason either player qualifies on their own.

Appendix D. Sensitivity: what changes and why

D1. Equal metric weights: exact bowling changes

The primary bowling model assigns 60% to bowler-creditable wickets and 40% to flagged boundaries conceded. Equal weighting changes this to 50/50. This increases the relative influence of boundary suppression and reduces the weight on wicket-taking; it does not mean the alternative is more correct. The 60/40 choice gives wickets the larger share because they are a direct dismissal outcome, while retaining a substantial 40% contribution for boundary control. No weights were fitted to maximise the observed Top 10.

Leaves primary Top 10	Enters under equal weights
Riley Meredith	Rasikh Salam
Adam Zampa	Sakib Hussain
Ben Dwarshuis	Spencer Johnson

The batter Top 10 remains 10/10 under equal weights. Under the alternative runs-and-wickets emphasis, Babar Azam leaves and Liam Livingstone enters; under the survival-and-boundary-control scenario, Glenn Phillips leaves and Matthew Wade enters. Exact alternative lists and scores are retained in `metric_weight_sensitivity.json`.

Alternative scenario	Batter change (out → in)	Bowler change (out → in)
Equal metric weights	None	Riley Meredith, Adam Zampa, Ben Dwarshuis → Rasikh Salam, Sakib Hussain, Spencer Johnson
Runs / wickets emphasis	Babar Azam → Liam Livingstone	Rahul Chahar → Mitchell Starc
Survival / boundary-control emphasis	Glenn Phillips → Matthew Wade	Riley Meredith, Adam Zampa, Ben Dwarshuis → Rasikh Salam, Sakib Hussain, Spencer Johnson
Boundary emphasis	None	Rahul Chahar → Mitchell Starc
Consistency heavier (40%)	None	Rahul Chahar → Rasikh Salam
Performance heavier (80%)	None	Nandre Burger → Sakib Hussain

D2. Missing boundary flags: deletion stress test

The source contains 23 records with recorded runs of 4 or 6 but neither boundary flag set (20 four-run records and 3 six-run records). Rather than silently infer the missing flags, the sensitivity test removes all 23 records from the analytical frame and fully recalculates peer baselines, reliability and rankings. This is deliberately a conservative whole-delivery deletion test, not a correction of the source fields.

Role	Top-10 membership overlap	Rank order unchanged	Interpretation
Batters	10/10	True	No selected batter changed position.
Bowlers	10/10	True	No selected bowler changed position.

Result: the primary Top 10 names and rank order are unchanged for both roles. This supports robustness to these 23 suspect rows, but it does not establish that every boundary flag in the dataset is correct.

D3. Weight and cutoff sensitivity

Equal metric weights retain 7/10 bowlers; the pressure-threshold alternatives retain 9/10 batters and 10/10 bowlers at an 8-over collapse cutoff, and 7/10 batters and 9/10 bowlers at 12 overs. These are material modelling sensitivities. The primary model is selected for interpretability and cricketing logic, not because it maximises overlap.

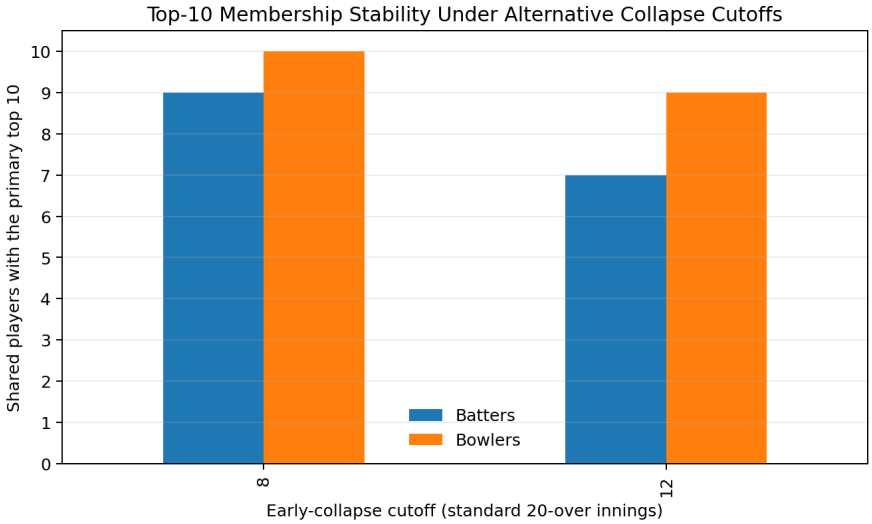


Figure D1. Top-10 overlap under alternative collapse cut-offs. The chart shows how many primary-list players remain when the early-collapse boundary moves; it does not show score quality or statistical significance.

Appendix E. Rajasthan Royals role-depth matrix

The matrix is a retrospective reading of the supplied IPL 2026 RR squad and observed phase usage. 'First choice', 'backup', 'injury cover' and 'successor watch' are planning labels based on observed usage and squad mapping; they are not an official 2027 depth chart, a fitness report or a retention prediction.

Role	First choice (2026 usage)	Backup	Injury cover	Successor watch
Top-order / powerplay batting	Vaibhav Sooryavanshi; Yashasvi Jaiswal	Dhruv Jurel	Lhuan-dre Pretorius; Ravi Singh	Aman Rao Perala
Indian middle-order batting	Dhruv Jurel; Riyan Parag	Ravindra Jadeja	Shubham Dubey; Ravi Singh	Aman Rao Perala
Finishing / lower-order batting	Donovan Ferreira; Ravindra Jadeja	Shubham Dubey; Shimron Hetmyer	Riyan Parag; Dasun Shanaka	Shubham Dubey; Ravi Singh
Wicketkeeper-batter	Dhruv Jurel	Donovan Ferreira	Lhuan-dre Pretorius; Ravi Singh	Aman Rao Perala
Death-overs pace	Jofra Archer; Brijesh Sharma	Sandeep Sharma; Nandre Burger; Tushar Deshpande	Adam Milne; Yudhvir Singh Charak	Brijesh Sharma; Kuldeep Sen
Powerplay pace	Jofra Archer; Nandre Burger	Brijesh Sharma; Sandeep Sharma; Tushar Deshpande	Adam Milne; Kwena Maphaka	Kuldeep Sen; Yudhvir Singh Charak
Middle-over spin	Ravindra Jadeja; Yash Punja	Ravi Bishnoi	Vignesh Puthur; Sushant Mishra	Vignesh Puthur; Yash Punja

E1. What the player rankings mean for RR

Player	Role-based reason	RR decision logic
Rinku Singh	Indian finisher profile; 605 pressure balls across 62 matches in the ranking dataset; index 81.71.	Potentially valuable because an Indian finisher preserves an overseas slot. Only assess if KKR releases him and the auction register confirms eligibility; price and role vacancy still matter.
Naman Dhir	Indian middle-order/utility profile; 211 pressure balls across 24 matches; index 82.69.	Could strengthen domestic middle-order depth behind/alongside Jurel and Parag. MI's release decision and RR's retained batting order must be checked first.
Ramandeep Singh	Indian batting all-rounder/lower-order profile; 106 pressure balls across 18 matches; index 75.19.	Offers lower-order flexibility rather than a like-for-like top-order bat. A depth option only if the retained all-rounder/finisher mix leaves a slot.
Indian death specialist	No Indian specialist fast/death bowler appears in the statistical bowling Top 10; Rahul Chahar is the only Indian bowler in that list and is a leg-spinner.	Do not infer RR lacks death bowling: its 2026 usage included Indian Brijesh, Sandeep and Tushar. The need is a succession/cover question after retention and fitness checks.

E2. Final conditional decision table

Priority	Decision	Evidence trigger before action
1	Protect pace/death coverage	Confirm retention, workload and fitness of Archer, Brijesh, Nandre, Sandeep and Tushar; only recruit if a real vacancy remains.
2	Maintain middle-over spin depth	Review Jadeja, Bishnoi and Yash Punja retention/development; no automatic spinner purchase.
3	Secure Indian finishing/middle-order depth	Evaluate Rinku, Naman or Ramandeep only if released, registered, affordable and role-compatible.
4	Do not treat model ranks as auction availability	As of 13 Nov 2026, retention deadline is reported as 17 Nov; final auction register is expected 3 Dec. Verify official lists before naming a purchasable target.

The IPL 2027 retention/release process is not yet complete at this report cut-off. The current reported retention deadline is 17 November 2026; the final auction register is expected on 3 December and the mini-auction is reported for 16 December in Goa. No ranked player is represented as currently available to RR.

Appendix F. Clean-run reproducibility and environment

This release includes the six original competition-supplied CSVs in `data/` so a reviewer can reproduce the analysis without locating a separate download. They are unmodified input files. Keep them within the competition/submission context and do not redistribute them publicly unless the organiser's terms explicitly permit it.

Clean-run commands

```
python -m pip install -r requirements.txt
python -m pytest -q
python main.py --audit-context --audit-pressure --audit-batting --audit-bowling --audit-baselines --audit-reliability
--audit-rankings --audit-pressure-ranking-stability --audit-rr-squad
python build_supplemental_outputs.py
python run_scope_comparison.py
python generate_report.py
```

Component	Version tested
Python	3.13.5
pandas	2.2.3
NumPy	2.3.5
Matplotlib	3.10.8
pytest	9.0.2
python-docx	1.2.0

The test suite passed 57 tests in the final environment. The machine-readable results are stored in `outputs/test_results.txt`; scope comparison metrics and player-level changes are stored in `outputs/scope_comparison/`.

The ranking CSVs remain authoritative for the primary order. The report tables and profile outputs are generated from those CSVs; no manual name substitution or score override is applied.